COMP3050 Individual Dissertation Single Honours

Prompt-engineering-based Hallucination Estimation for large language models

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April 9, 2026

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Abstract

Large Language Models (LLMs) have achieved remarkable performance across a wide range of natural language processing tasks. However, LLMs often generate factually incorrect or fabricated information, a phenomenon commonly referred to as hallucination. While existing research has primarily focused on hallucination detection, effective mitigation strategies—particularly in open-domain question answering settings where explicit grounding is limited—remain challenging.

This project presents a multi-signal framework for hallucination detection and estimation, coupled with detector-guided mitigation strategies, including re-prompting, candidate filtering, and abstention. The proposed system integrates multiple complementary signals, including LLM-as-a-judge evaluation, natural language inference (NLI) for factual consistency, and SelfCheck-based response consistency, to produce a unified hallucination score. Based on this score, a detector-conditioned re-prompting strategy is applied to selectively revise responses or abstain when confidence is low. The framework is evaluated on multiple benchmark datasets, including TruthfulQA, FEVER, and Natural Questions, using both performance and stability metrics.

Experimental results show that the approach is most effective on evidence-grounded datasets such as FEVER, while gains on more open-ended benchmarks are limited. In particular, notable reductions in hallucination rates are observed on evidence-based datasets such as FEVER. However, performance improvements are less consistent on open-ended datasets, highlighting the challenges of mitigating hallucinations in the absence of explicit grounding.

Overall, this work highlights the importance of integrating multiple evaluation signals with adaptive mitigation strategies and shows that the effectiveness of hallucination mitigation is strongly dependent on the availability of grounding evidence.

Contents

Chapter 1

Introduction

1.1 Background

In recent years, large language models (LLMs) have demonstrated outstanding performance in various natural language processing tasks, including question answering, text summarization, language inference, knowledge retrieval, and dialogue generation [1]. These models are capable of generating fluent and context-coherent responses, thus finding widespread application in research and practical applications, such as clinical decision support systems [2]. Despite these advances, LLMs are known to generate factually incorrect or fabricated information that is not supported by the input or external knowledge, a phenomenon often referred to as "hallucination" [3]. These hallucinations can occur even when model outputs appear linguistically plausible, making them difficult to detect and potentially misleading to users. As a result, they can degrade system performance and fail to meet user expectations [4].

1.2 Motivation

Hallucinations in large language models pose a serious challenge to their reliability and practical deployment. In real-world applications, users often perceive model outputs as reliable sources of information, making hallucinated outputs highly misleading in real-world use. These errors can lead not only to distorted information dissemination, decreased user trust, and reduced system performance, but also to erroneous task execution in LLM-based agent systems, further impacting overall system reliability [5]. In high-risk applications such as medical decision support, while LLM-assisted tools can effectively support goal elicitation, clarification, and revision, the lack of human interaction, personalized patient contextual information, and the significant risk of overload due to unconstrained information access pose significant challenges [6]. These issues can amplify uncertainty and potential harm, limiting their safe application as independent asynchronous tools in real-world scenarios.

The causes of hallucinations are multifaceted, including both data-related and model-related factors. Training data may contain quality issues, including noise, bias, outdated information, subjective content, and unverified facts, leading to inaccurate or unsubstantiated model outputs [7]. Furthermore, LLMs are often poorly calibrated and prone to overconfidence, assigning excessively high confidence levels to erroneous outputs [8]. In open-domain environments, this problem is further exacerbated by a lack of reliable underlying information, allowing models to generate seemingly plausible but actually fabricated explanations [9]. These challenges collectively underscore the need for more robust methods to reliably detect, estimate and mitigate hallucinations.

Despite the growing interest in hallucination detection, existing methods remain limited in several aspects. Many hallucination evaluation methods rely on single-model or single-signal strategies, such as confidence estimation, natural language inference (NLI), or response consistency checks [4]. While these approaches provide advantages in handling lexical variations or detecting simple inconsistencies, they are prone to error propagation from underlying models, struggle to pinpoint hallucinated substrings at a fine-grained level, and often fail to fully capture the diversity (e.g., intrinsic vs. extrinsic) and complexity of hallucinations [4]. Therefore, these methods are often unstable and lack robustness across different tasks and datasets.

Furthermore, while detection methods have been extensively studied, effective mitigation strategies remain relatively scarce. Existing mitigation methods are often heuristic or poorly integrated with detection mechanisms, limiting their effectiveness in reducing hallucinatory outputs. This challenge is particularly pronounced in knowledge-intensive task environments such as open-domain question answering, because reliable contextual information is limited and pre-trained language models struggle to access and manipulate knowledge accurately, are prone to hallucinations, and fail to provide reliable sources for predictions [9]. These limitations underscore the need to develop a more comprehensive and integrated framework that combines detection, estimation, and mitigation measures to improve the factual reliability of LLM-generated outputs.

1.3 Research Gap

Despite recent progress in hallucination detection and mitigation, several key limitations remain. Firstly, many existing hallucination detection methods tend to rely on a single signal, such as model uncertainty estimation, Natural Language Entailment (NLI), or consistency among sampled responses. This dependence on a single source of information limits their robustness across different tasks, models, and datasets [10]. Secondly, research on hallucination detection metrics and mitigation strategies has largely proceeded in parallel. The decoupling of these two processes results in a lack of cohesive intervention strategies, where detection results are not effectively integrated into the generation loop to facilitate real-time mitigation [4]. Thirdly, existing hallucination evaluation metrics and mitigation methods often exhibit limited cross-dataset generalizability. While performing well on specific benchmarks, these approaches struggle to transfer effectively to other datasets or unseen domains [4].

These limitations highlight the need for more integrated and robust approaches. In particular, there is a need for multi-signal frameworks that not only improve the reliability of hallucination detection but also enable effective, detector-guided mitigation. To address this gap, this project proposes a unified framework that integrates multi-signal evaluation with detector-conditioned intervention strategies, including re-prompting, candidate filtering, and abstention mechanisms.

1.4 Aims and Objectives

This project aims to develop a unified and robust framework for hallucination detection, estimation, and mitigation in large language models (LLMs). Specifically, it addresses the limitations of existing single-signal methods and investigates how the effectiveness of hallucination mitigation varies across tasks with different levels of grounding availability. To achieve this goal, the project follows these objectives:

- To design a multi-signal hallucination detection mechanism that integrates complementary evaluation signals, including:

    1. an LLM-as-a-judge evaluation component
    2. a SelfCheckGPT-style response consistency layer based on contradiction esti-

mation across multiple generated outputs

3. an entailment-based fact verification module using natural language inference (NLI)

- To develop a quantitative hallucination estimation method that aggregates multiple evaluation signals and provides stable and reproducible measurements, while enabling the analysis of variability and consistency across different datasets through repeated sampling and statistical analysis (e.g., mean and variance).

- To implement a detector-conditioned mitigation strategy that forms a closed-loop pipeline, incorporating techniques such as re-prompting, candidate filtering, and abstention, with response selection guided by hallucination risk scores.

- To evaluate the proposed framework across multiple benchmark datasets, including TruthfulQA, FEVER, and Natural Questions, using performance and stability metrics, and to analyse dataset-dependent performance and the role of grounding.

- To conduct ablation studies and threshold tuning to assess the contribution of individual components and optimise the overall performance of the proposed framework.

1.5 Contributions

This project makes the following contributions:

- It proposes a unified multi-signal framework that integrates complementary evaluation signals for hallucination detection, estimation, and mitigation, enhancing robustness and stability compared to single-signal approaches, while revealing trade-offs between accuracy and consistency.

- It introduces a quantitative and stability-aware hallucination estimation method that leverages repeated sampling and statistical analysis to produce more reliable and reproducible measurements.

- It develops a detector-conditioned mitigation strategy that forms a closed-loop pipeline, enabling dynamic response revision, filtering, and selection based on hallucination risk.

- It provides a comprehensive cross-dataset evaluation on TruthfulQA, FEVER, and Natural Questions, demonstrating the effectiveness and limitations of the proposed approach in both grounded and open-domain settings.

- It offers empirical insights into the behaviour of multi-signal hallucination detection, including component contributions, stability characteristics, and challenges in open-domain scenarios.

Chapter 2

Literature Review

This chapter provides an overview of existing research related to hallucination in large language models, with a particular focus on detection, estimation, and mitigation approaches. The aim is to analyse the strengths and limitations of current methods and to identify gaps that motivate the proposed framework. This review is structured as follows: Section 2.1 introduces the concept and types of hallucinations in large language models. Section 2.2 examines the underlying causes of hallucination, including both data-related and model-related factors. Section 2.3 examines existing hallucination detection methods, including confidence-based, entailment-based, and consistency-based approaches. Section 2.4 discusses hallucination estimation and uncertainty quantification techniques. Section 2.5 presents mitigation strategies, including prompt-based and retrieval-augmented approaches. Section 2.6 highlights reproducibility challenges in hallucination research. Finally, Section 2.7 summarises the key limitations of existing work and identifies the research gap addressed in this project.

2.1 Concept and Types of Hallucination in LLMs

Hallucinations in Large Language Models (LLMs) refer to the generation of content that is factually inaccurate or contains fabricated information. While LLMs can generate fluent and contextually coherent responses, such outputs may lack factual grounding. These errors are difficult to detect, mislead users, and potentially undermine their reliability [3].

Existing literature has proposed different classification methods to describe hallucination phenomena. One widely accepted method is to divide hallucinations into intrinsic and extrinsic types [11]. Intrinsic hallucinations occur when the generated content contradicts the provided input or context [12]. Conversely, extrinsic hallucinations occur when the model introduces external information that the input does not support, and this information may be fabricated or unverifiable [12]. This distinction highlights that hallucinations can stem from inconsistencies with the input or from a lack of reliable evidence. In addition, hallucinations can also be divided into factuality hallucinations and faithfulness hallucinations. Factuality hallucinations emphasize that the generated content is inconsistent with real-world facts, even if it doesn't contradict the input [11]. In contrast, faithfulness hallucinations refer to generated content lacking support from the given input or source context, or being inconsistent with it [11]. This perspective emphasizes that hallucinations can stem from a lack of factual accuracy or a failure to faithfully follow the provided input, further illustrating the multifaceted nature of hallucinations. Beyond this binary classification, hallucinations can also be understood in relation to knowledge grounding and verification. In knowledge-intensive tasks, such as open-domain question answering, the lack of explicit external evidence often leads to hallucinations, making it difficult to verify the correctness of generated answers. This further increases the risk of producing seemingly reasonable but factually incorrect outputs [13].

In summary, hallucination in language generation models large language models (LLMs) is a multifaceted phenomenon stemming from limitations in language generation and the knowledge grounding base. This complexity necessitates robust methods for detecting, estimating, and mitigating hallucination, which will be explored in the following chapters.

2.2 Causes of Hallucination

Hallucinations in large language models (LLMs) arise from a combination of data-related, model-related, and task-specific factors. These factors interact in complex ways, contributing to the generation of outputs that are fluent but factually incorrect.

From a data perspective, source-reference divergence in training data is a significant cause of hallucinations. When the target text contains information not supported by the input, the model learns generation patterns that do not fully rely on the source text [4]. This bias often stems from heuristic data collection processes or inherent characteristics of certain natural language generation tasks. Furthermore, large-scale training corpora may contain noise, biases, duplicate content, or unverified information. Consequently, the model may internalize and reproduce these defects, leading to inaccurate or unsubstantiated outputs [4]. These data-related issues collectively contribute to the generation of hallucinatory content in natural language models. More broadly, they reflect underlying data quality challenges, where inconsistencies between source and target, as well as noisy or biased training corpora, can systematically influence model behaviour.

From a modeling perspective, Large Language Models (LLMs) typically suffer from poor calibration and are prone to overconfidence, assigning excessively high confidence levels to incorrect predictions [14]. This phenomenon is closely related to the model's autoregressive training objective. LLMs primarily generate output by predicting the next most likely token, which makes the model more inclined during training to generate fluent, coherent, and confident outputs, rather than rigorously ensuring factual accuracy or estimating inherent uncertainty [15].

From a task and environment perspective, hallucinations are particularly prevalent in open-domain and knowledge-intensive settings. In these scenarios, he model primarily relies on parametric knowledge learned during pre-training and cannot acquire explicitly provided supporting evidence. The lack of reliable external evidence makes it extremely difficult to verify the correctness of the generated answers, which significantly increases the risk of producing seemingly reasonable but actually wrong answers.[16]. This challenge is even more pronounced in long-form content generation tasks, as maintaining factual consistency becomes increasingly difficult with longer outputs [17].

Overall, these factors not only explain the prevalence of hallucination, but also highlight the limitations of existing approaches that rely on single-source verification or surface-level evaluation signals. Hallucinations in LLMs result from the combined influence of imperfect training data, limitations in model calibration and objectives, and the inherent challenges of knowledge-intensive tasks. These insights motivate the need for more robust and integrated approaches to effectively detect, estimate, and mitigate hallucinations, as discussed in the following sections.

2.3 Hallucination Detection Methods

A wide range of approaches has been proposed to detect hallucinations in large language models (LLMs). Existing methods can be broadly categorised based on the source of information used for evaluation, including internal signal-based methods, external verification-based methods, and model-based evaluation methods. Internal signal-based methods rely on the model's own uncertainty or output variability, such as uncertainty-based and consistency-based approaches. External verification-based methods assess factual alignment using external evidence, including entailment-based and retrieval-based approaches. In addition, model-based evaluation methods utilise large language models as evaluators, commonly referred to as LLM-as-a-judge methods. While these approaches differ in their underlying assumptions, required resources, and robustness across tasks, each captures only partial aspects of hallucination. This section reviews these methods and provides a critical analysis of their strengths and limitations.

2.3.1 Internal signal-based methods

Uncertainty-based methods

Uncertainty-based methods detect hallucinations by estimating the model's uncertainty in its generated outputs. A common approach is to analyse internal probability signals, such as token-level probabilities, conditional sequence probabilities (log-likelihood), or entropy measures, under the assumption that higher uncertainty (or lower confidence) is associated with an increased risk of unreliable or hallucinated content [18].

In practice, these methods are widely adopted due to their simplicity and computational efficiency. They are typically unsupervised and training-free, relying solely on the model's internal output, such as probability estimation or multiple sampling, without requiring additional models, external knowledge bases, or retrieval mechanisms [18]. Therefore, they provide a lightweight and practical solution for hallucination detection, especially suitable for resource-constrained environments.

However, uncertainty estimation based solely on model confidence is fundamentally limited by the poor calibration of large language models. Prior research has shown that large language models (LLMs) often exhibit overconfidence, producing overconfident, plausible falsehoods by assigning high probabilities to incorrect or unsupported outputs [19]. As a result, the internal confidence signal cannot reliably reflect the correctness of the facts, making overconfident hallucinations difficult to detect where hallucinatory content is given high confidence. Furthermore, such methods rely purely on internal signals and do not incorporate external evidence or semantic consistency, limiting their effectiveness in complex or knowledge-intensive tasks.

Therefore, although uncertainty methods are lightweight and do not require training, their reliance on internal confidence limits their reliability, and they usually need to be combined with external validation or model evaluation methods to improve their robustness in complex tasks.

Consistency-based methods

Consistency-based methods also detect hallucinations by analyzing the consistency of multiple sampled responses generated by a model for the same input. The basic assumption is that when the model possesses true knowledge, different sampled responses tend to produce similar and factually consistent content; however, for hallucinatory content, different sampled responses are more likely to diverge or even contradict each other [10].

In practice, these consistency-based methods typically generate multiple candidate responses or inference paths by stochastic sampling of the same input and then evaluate their consistency. Consistency can be quantified through semantic similarity, contradiction detection, natural language inference (NLI), or answer aggregation. For example, SelfCheckGPT [10] estimates the risk of hallucination by comparing multiple sampled responses generated by the model for the same query and identifying information contradictions or divergences, without requiring any external knowledge base or internal model probability distribution. Similarly, Self-Consistency [20] significantly improves inference accuracy in chain-of-thought inference by sampling multiple different inference paths and selecting the most frequent (e.g., most consistent) answer in the final response.

These methods collectively embody the idea of unsupervised, zero-resource consistency detection, making it possible to detect semantic inconsistencies through cross-sampling without the need for additional supervisory signals or external resources.

However, consistency-based methods also have certain limitations. First, these methods typically require multiple sampling and forward propagation steps, increasing computational cost and inference latency [4]. Second, these methods implicitly assume that the correct answer is consistent across different generated results, but this assumption may not hold when multiple reasonable or equivalent answers exist [21]. Finally, output consistency does not necessarily imply factual correctness, as the model may repeatedly generate consistent but erroneous information due to biases in the training data or parameterized knowledge [22].

Therefore, while consistency-based methods offer a lightweight and unsupervised approach to hallucination detection, they typically require the combination of external validation or model-based evaluation techniques to improve their reliability in complex and knowledge-intensive tasks.

2.3.2 External / verification-based methods

Entailment-based methods (NLI)

Entailment-based methods use Natural Language Inference (NLI) to verify whether generated content is logically supported by references or evidence, thereby detecting hallucinations. The core idea is to model hallucination detection as a textual implication problem: if the generated answer is implied by the reference text, it is considered reliable; if it contradicts the reference text or lacks supporting evidence, it is considered a hallucination [23].

In practice, these methods typically employ pre-trained Natural Language Inference (NLI) models (e.g., models trained on MultiNLI) to compute the entailment, contradiction, and neutral probabilities between the reference text and the generated answer [24]. Entailment probability is commonly used as a primary indicator of factual consistency, while contradiction signals are used to identify and penalise unsupported or incorrect content [25]. Prior research has demonstrated that NLI methods adapted with appropriate granularity can effectively detect factual inconsistencies in text summarization and achieve strong performance across multiple benchmarks [26].

By modeling factual consistency as a semantic alignment problem between generated content and reference evidence, such methods can be applied to a variety of tasks and exhibit strong generalization capabilities [27]. A key advantage of entailment-based methods is their ability to go beyond surface matching and incorporate semantic reasoning, thus enabling more robust detection of factual inconsistencies. Unlike uncertainty-based methods or consistency-based methods, Natural Language Inference (NLI) explicitly evaluates the relationship between generated content and referenced knowledge, making it particularly effective in knowledge-based tasks.

However, these methods also have certain limitations. First, their performance largely depends on the quality of the references or evidence, which in real-world applications are often incomplete, noisy, ambiguous, or lack coverage, thus affecting the reliability of consistency assessments [23]. Second, NLI models themselves are not perfectly accurate and may produce erroneous implication judgments in complex contexts or long text scenarios, introducing additional prediction bias [26]. Furthermore, these methods typically require additional computational resources (e.g., multiple inferences or calls to external models) and dependence on external reference or retrieval systems, increasing computational overhead and limiting their scalability in large-scale or open environments [26].

Therefore, while entailment-based methods provide a principled and semantically grounded approach to hallucination detection, they are often combined with other signals, such as uncertainty estimation or consistency-based methods, to achieve more reliable and robust performance in practice.

Retrieval-based methods

Retrieval-based methods detect hallucinations by associating generated content with external knowledge sources, such as knowledge bases, documents, or online corpora. The core idea is to verify whether the model's output is supported by retrieved evidence, thereby reducing reliance on knowledge of model parameters and improving factual reliability [9].

In practice, these methods typically begin by retrieving relevant documents or paragraphs using information retrieval techniques (such as BM25 or dense retrieval), and then align and verify the generated answer with the retrieved evidence [28]. Specifically, retrieval-based methods employ various evaluation strategies to measure their consistency, such as semantic similarity-based matching methods and model-based evaluation metrics, such as language model-based discriminative methods [29]. Depending on the verification method, these methods can be further divided into two categories. One category involves introducing a retrieval mechanism during the generation stage, such as the Retrieval-Augmented Generation (RAG) framework, which directly generates evidence-based output by combining retrieval and generation [9]. The other category is to perform post-hoc verification after generation, which involves explicitly checking whether the generated content can be supported by external knowledge sources [30]. The key advantage of retrieval-based methods lies in their ability to incorporate up-to-date external knowledge, which improves performance in open-domain and knowledge-intensive tasks. By grounding generated content in retrieved evidence, these methods can effectively reduce hallucinations and enhance interpretability.

However, retrieval-based methods also have certain limitations. First, their performance is highly dependent on the quality of the retrieval component. If relevant evidence is not retrieved, the system may fail to generate a correct answer [31]. Second, retrieval systems may introduce additional noise or irrelevant information (i.e., unhelpful content for the target task), which can exert negative impacts on the subsequent verification process [32]. Furthermore, these methods typically require complex processes involving retrieval, ranking, and verification phases, increasing system complexity and computational cost [9]. Finally, accessing external knowledge sources is not always feasible, especially in resource-constrained situations or when real-time processing is required [33].

Therefore, while retrieval-based methods provide strong external grounding for hallucination detection, they are often combined with other approaches, such as entailment-based or consistency-based methods, to achieve more robust and reliable performance in practice.

2.3.3 Model-based evaluation methods

LLM-as-a-Judge methods

LLM-as-a-Judge methods utilize large language models themselves as evaluators to assess the authenticity of generated content, thereby detecting hallucinations. These methods do not rely solely on internal confidence signals or external validation, but rather guide the LLM to explicitly determine whether a given answer is factual, consistent, and supported by evidence [34]. This approach redefines hallucination detection as a model-based evaluation problem, where the model acts as a reasoning-based critic.

In practice, the LLM-as-a-Judge method typically guides the language model's evaluation by constructing evaluation prompts, using the input question, generated answer, and optional reference answers or supporting evidence as context [35]. Subsequently, the LLM generates discrete judgments (e.g., true/false, supported/unsupported) or continuous scores to reflect its factual consistency [35]. This approach models hallucination detection as a reasoning-based evaluation task, enabling the model to leverage its implicit knowledge and reasoning abilities to make judgments. Prior research has shown that large language models can serve as effective evaluators, exhibiting high consistency with human judgments across various evaluation tasks [34]. The key advantage of large language models as evaluators lies in their flexibility and versatility. Unlike uncertainty-based methods, they do not rely on internal probabilistic signals. Moreover, unlike retrieval-based methods, they do not require explicit access to external knowledge sources. Instead, they leverage the implicit knowledge and reasoning capabilities of large language models, enabling them to be applied to a wide range of tasks without additional training or task-specific design [36].

However, these methods also have significant limitations. First, the reliability of the evaluation depends on the capability and calibration of the evaluation model itself, which may still be biased or prone to hallucinations [37]. Existing research has shown that LLMs exhibit significant positional bias when used as evaluators; simply changing the answer order can lead to a high proportion of evaluation inconsistencies, even in high-performance models such as GPT-4[37]. Second, evaluation results may be sensitive to prompting design and may lack stability under different prompt or sampling strategies. Research shows that LLMs are highly sensitive to prompting design; the prompt itself may become a target for attacks or lead to unstable results [38]. In addition, research has shown that even under deterministic settings (e.g., fixed seed and temperature ≈ 0), LLM outputs remain stochastic samples from the underlying probability distribution, exhibiting so-called "fixed randomness". As a result, minor changes such as random seed variation can lead to substantial fluctuations in evaluation consistency, and temperature settings can further affect evaluation reliabilityly [39]. Third, using a large language model as the evaluator increases computational costs, especially when multiple evaluations are required [40].

Therefore, while LLM-as-a-Judge methods provide a flexible and scalable approach to hallucination detection, they are often combined with other signals, such as entailment-based verification or consistency-based methods, to improve robustness and reliability in practice. This motivates the need for combining LLM-based evaluation with external verification or consistency-based signals, as adopted in this work.

2.4 Hallucination Mitigation Methods

Hallucination mitigation methods aim to reduce the generation of factually incorrect or unsupported content by large language models, complementing detection approaches by actively improving output reliability. Unlike detection methods, which identify hallucinations after generation, mitigation methods focus on preventing or correcting such errors during or after the generation process.

2.4.1 Prompt-based approaches

Prompt-based mitigation methods aim to reduce hallucinations by influencing the model's generative behavior through carefully crafted input instructions. Research has shown that using well-designed few-shot prompts can significantly improve model performance without modifying model parameters [41]. These methods rely on prompt design to encourage the model to respond more cautiously, reasoning-oriented, and in an evidence-based manner. Existing research indicates that uncertainty-aware prompting strategies, such as allowing the model to abstain or avoid generating unproven assertions, can effectively reduce the incidence of hallucinations by promoting conservative generation [8]. In particular, prompting techniques that guide the model to make stepwise inferences or explicitly consider evidence, such as Chain-of-Thought reasoning, have been shown to improve factual consistency and reliability across various tasks. Prior work shows that providing intermediate reasoning steps in prompts (e.g., Chain-of-Thought) enables large language models to decompose complex problems into sub-steps, thereby improving accuracy across tasks such as arithmetic, commonsense, and symbolic reasoning [36].

However, prompt-based methods are inherently heuristic and highly sensitive to prompt design, often exhibiting significant performance variations across different models and tasks. Specifically, the same prompt can result in vastly different performances across different models, and the performance of the same model can also fluctuate significantly across different tasks [38]. Furthermore, these methods do not fundamentally address the mechanisms underlying hallucination generation, as the model still relies on its internal parametric knowledge for generation [21]. Even with prompt guidance, the model's generation remains constrained by its training objective (predicting human text) rather than pursuing factual correctness[21].

As a result, prompt-based mitigation is often combined with other approaches to achieve more robust performance. This highlights the limitation of prompt-based approaches as a lightweight but fundamentally incomplete solution to hallucination mitigation.

2.4.2 Retrieval-augmented approaches

Retrieval-augmented approaches aim to mitigate the uncertainty of model predictions by grounding model outputs in external knowledge sources, such as documents, knowledge bases, or online corpora. Instead of relying solely on parametric knowledge, these methods incorporate retrieved evidence to improve factual accuracy and reliability [9]. In practice, retrieval-augmented methods usually consist of two main stages: (1) using sparse or dense retrieval techniques (such as BM25 or neural retrievers like DPR) to retrieve relevant documents; and (2) inputting the retrieved context into a sequence-to-sequence model for conditional generation. By explicitly integrating external evidence, these methods reduce reliance on potentially outdated or incomplete parametric knowledge and significantly improve factual consistency, especially in knowledge-intensive tasks [28]. Existing research shows that retrieval-augmented models can significantly improve the performance of knowledge-intensive tasks such as open-domain question answering and fact verification, in which the acquisition of external knowledge is crucial [9], [42]. In particular, building the generation process on the retrieved evidence can not only effectively reduce model hallucinations, but also enhance the interpretability of the output, because the generated answers can be directly traced back to specific supporting documents [28], [33].

However, retrieval-augmented approaches also exhibit several limitations. First of all, their effectiveness is highly dependent on the retrieval component and the relevance of the evidence retrieved. If the fully relevant evidence is not successfully retrieved, the model may still produce unfounded or incorrect output, and even hallucinations [4], [9], [28]. Secondly, the retrieval system may introduce noise or irrelevant information, which can interfere with the generator's judgment and negatively impact output quality. This low-quality context may cause the model to perform incorrect logical reasoning and even create conflicts between existing evidence and its parametric knowledge [9], [43]. Thirdly, these methods typically involve multi-stage pipelines (retrieval, ranking, and generation), increasing system complexity and computational cost. These limitations highlight that the performance of retrieval-augmented methods is closely tied to the availability and quality of grounding information.

Overall, retrieval-augmented approaches provide a powerful mechanism for reducing hallucinations by introducing external grounding. However, their reliance on retrieval quality and external knowledge availability motivates the need for complementary approaches, such as multi-signal evaluation and detector-guided mitigation, to achieve more robust performance across diverse tasks.

2.4.3 Post-hoc correction methods

Post-hoc correction methods aim to reduce hallucinations by iteratively improving or correcting the model output after the initial generation. Unlike prompt-based or retrieval-augmented methods, these methods do not directly intervene in the generation process but introduce additional feedback and editing steps to improve the quality and factual consistency of the generated response. For example, Self-Refine [44] uses the same language model to first generate an initial output, then produce feedback on that output, and iteratively refine it, significantly improving output quality without additional training. In practice, post-hoc correction methods typically adopt a multi-stage process: the language model first generates an initial response, which is then iteratively improved through evaluation and modification steps. A mainstream strategy is self-correction, which prompts the same model to re-evaluate its output and identify potential factual errors, inconsistencies, or logical issues. This process often combines external feedback signals (e.g., task success metrics or unit test results) for consistency checks with internal reasoning to generate reflective feedback, ultimately enabling explicit verification and correction of the output [10], [44], [45]. Existing research shows that iterative improvement and self-correction mechanisms can help models detect and correct their own errors, thereby significantly improving the factual reliability of the output. In particular, Self-Refine [44] employs self-feedback and iterative refinement, while SelfCheckGPT [10] uses multi-sampling consistency checks and internal feedback loops to identify inconsistencies and reduce hallucinations. These methods are especially effective when external knowledge is limited or lacking, as they primarily rely on the model's own internal reasoning ability and multi-round self-assessment process.

However, post-hoc correction methods also have several important limitations. First, their effectiveness is essentially limited by the capabilities of the underlying model, as both generation and error correction depend on the same or similar model representations and reasoning patterns, which can easily lead to the retention or even reinforcement of systematic errors [44], [45]. In addition, such methods usually incur high computational overhead, and the quality of self-feedback is difficult to guarantee when external knowledge is absent or the base model is weak [10]. These limitations highlight the need to develop a more robust multi-signal detection and intervention framework.

Overall, post-hoc correction methods provide a flexible and model-agnostic approach to improving factual consistency by introducing an additional verification stage after generation. However, their reliance on internal model capabilities and iterative refinement highlights the need for complementary approaches, such as external grounding and multi-signal evaluation, to achieve more robust hallucination mitigation.

2.4.4 Training-based methods (RLHF / fine-tuning)

Training-based methods aim to mitigate hallucinations by directly modifying the model's behavioral patterns through additional training procedures, such as Supervised Fine-Tuning (SFT) or Reinforcement Learning from Human Feedback (RLHF). These approaches influence the model at the parameter level, enabling it to generate more reliable and aligned responses during inference [41]. In practice, supervised fine-tuning (SFT) involves training the model on high-quality, carefully curated datasets of human demonstrations that provide correct and factually grounded responses [41]. This process enables the model to learn more accurate input-output mappings and substantially reduces the likelihood of generating unsubstantiated or erroneous content. Furthermore, reinforcement learning from human feedback (RLHF) incorporates human preferences into the optimization process by training a reward model [41], [46], thereby further refining model behavior and guiding it to produce more helpful, truthful, and human-aligned responses. Existing research demonstrates that training-based alignment techniques can significantly enhance model response quality and effectively reduce harmful or misleading outputs [41], [47]. In particular, RLHF has been widely adopted in the development of modern large language models, improving alignment and controllability, which substantially increases the truthfulness and factual reliability of outputs and thereby fosters greater user trust. These methods are particularly effective at shaping the model's general behavior across diverse tasks because they directly optimise model parameters, rather than relying solely on external retrieval or post-hoc interventions during inference [41].

However, training-based methods also have several important limitations. First, alignment achieved through fine-tuning or RLHF cannot fully guarantee factual correctness; the model may generate responses that appear plausible but are factually incorrect in order to maximize the reward signal, resulting in so-called "alignment without truthfulness" [41]. Second, these methods are highly dependent on the quality of training data, where annotator biases or inaccuracies can be amplified during the fine-tuning process [47]. Third, training-based approaches typically require substantial computational resources and high human annotation costs, which limit their scalability in large-scale applications [41]. These limitations indicate that parameter-level alignment alone is insufficient to eliminate hallucinations, underscoring the urgent need to incorporate external validation signals as a complement.

Overall, training-based methods provide a powerful mechanism for improving model behaviour by directly shaping its internal representations. However, their limitations highlight that alignment alone is insufficient to fully address hallucinations, motivating the integration of complementary approaches such as external grounding and multi-signal evaluation for more robust hallucination mitigation.

2.5 Reproducibility Challenges in Hallucination Research

For large language models (LLMs), reproducibility has emerged as a critical challenge in hallucination research. Although deterministic decoding settings (e.g., fixed temperature and random seed) are commonly employed, the inherent stochasticity of the underlying generation process can still lead to significant output variability. Consequently, repeated runs of the same model on identical inputs may produce different responses, resulting in inconsistency in hallucination detection and evaluation [10]. A primary source of this variability lies in the probabilistic nature of language generation. Even under constrained decoding conditions, the outputs of Large Language Models (LLMs) are sampled from a probability distribution over possible continuations. Consequently, minor perturbations, such as changes in the sampling seed, adjustments to decoding parameters, or even subtle fluctuations in the underlying computational environment, can lead to significant discrepancies in both the generated content and downstream evaluation results [38]. This inherent uncertainty arising from probabilistic sampling contributes to the model's diminished robustness, particularly when processing edge cases or out-of-distribution inputs.

Furthermore, hallucination detection methods often exhibit instability. Prompt-based evaluation approaches are highly sensitive to prompt engineering, where minor changes in wording or format can significantly alter the model's judgment. Moreover, LLM-as-a-judge methods are frequently affected by position bias, verbosity bias (favoring longer answers), and scoring inconsistencies, further weakening the reliability of the evaluation results [34]. These issues are exacerbated by cross-dataset variability: detection algorithms that perform well on evidence-retrieval tasks often struggle to generalize effectively to more adversarial or open-ended tasks, reflecting the limited robustness of current hallucination detection approaches across diverse evaluation settings.

Motivated by these limitations, this project incorporates a stability-aware evaluation framework that complements traditional accuracy metrics with statistical measures such as mean performance and variance across repeated runs. This enables a more robust and reproducible assessment of hallucination detection and mitigation performance across different datasets and experimental settings.

2.6 Summary

In summary, existing research has explored a wide range of approaches for hallucination detection and mitigation in large language models, including prompt-based methods, retrieval-augmented generation, post-hoc correction, and training-based alignment. While these methods have demonstrated effectiveness under specific conditions, they also exhibit notable limitations in terms of robustness, stability, and generalization across tasks and datasets.

In particular, many approaches rely on single evaluation signals, and detection and mitigation are often treated as separate processes. Moreover, as discussed in Section 2.5, hallucination evaluation itself is inherently unstable due to stochasticity, prompt sensitivity, and model biases. These issues are further amplified in cross-dataset settings, where performance varies significantly between evidence-grounded and open-domain tasks.

These limitations motivate the need for more integrated and robust approaches to hallucination detection and mitigation, which is the focus of this project.

Chapter 3

Research Gap and Design Motivation

Despite the wide range of hallucination detection and mitigation approaches discussed in Chapter 2.6, several fundamental limitations remain.

First, many existing detection methods rely on a single signal (e.g., uncertainty, entailment, or self-consistency), which limits their robustness across different tasks and datasets. As shown in prior work, these signals often capture only partial aspects of factual correctness and may fail under distribution shifts.

Second, hallucination detection and mitigation are typically treated as separate processes. Detection methods aim to identify errors after generation, while mitigation approaches focus on improving generation quality. However, the lack of interaction between these two stages limits the overall effectiveness of hallucination reduction.

Third, existing approaches suffer from instability and reproducibility issues. As discussed in Section 2.5, even under controlled settings, LLM outputs exhibit stochastic variability, leading to inconsistent evaluation results. This instability is further amplified in prompt-based evaluation and LLM-as-a-judge methods.

Finally, many mitigation approaches either rely heavily on external knowledge (e.g., retrieval-augmented methods) or internal self-correction, both of which have inherent limitations. External retrieval depends on evidence quality, while self-correction is constrained by the model's own reasoning ability.

Motivated by these limitations, this project proposes a unified multi-signal hallucination detection and closed-loop mitigation framework, which integrates semantic judgement, NLI-based verification, and self-consistency signals, while explicitly addressing stability and cross-dataset generalisation.

Chapter 4

Methodology and System design

This chapter introduces the method framework and system design developed for hallucination detection and mitigation in large language models. Existing hallucination detection methods usually rely on a single signal, such as model confidence, entailment scores, or self-consistency. However, these methods have some well-known limitations, including instability under different prompts, sensitivity to model bias, and poor effectiveness in the case of logical coherence but reasoning errors. In addition, many previous works regard hallucination detection as a post-hoc evaluation task without integrating it into the generation process. In order to meet these challenges, this project proposes a closed-loop hallucination detection, estimation, and mitigation framework. The framework combines a variety of complementary signals to evaluate the factual reliability. Specifically, the system integrates semantic judgment, NLI-based factual consistency, and self-consistency analysis into a unified scoring mechanism. The score is then used to trigger threshold-based mitigation strategies, including answer rewriting, candidate answer filtering, reranking, and abstention, so that the system can actively correct unreliable output. The proposed framework is intended to be independent of the dataset, and only a small amount of adjustment is required to adapt to different evaluation settings and answer formats. This chapter focuses on the technical design and implementation of the proposed unified framework, including the formalization of the problem, the construction of the multi-signal hallucination detection model and the design of the detector-conditioned mitigation mechanism. In addition, this chapter also introduces the threshold setting and calibration procedures used to convert test scores into actionable decision-making, as well as the evaluation protocols used to ensure the robustness and repeatability of the experimental results. Together, these components form a coherent system aimed at reducing hallucinations and maintaining the quality of answers.

4.1 Problem Formulation

This project aims to address the problem of hallucination detection, estimation, and mitigation in large language models (LLMs) through a prompt-engineering-based framework. Given a claim (or question), an optional evidence context (if available), and a model-generated answer, the objective is to assess the factual reliability of the answer, both in terms of binary classification and continuous estimation, and to apply corrective actions when necessary to reduce hallucination.

Formally, let q denote the input claim (or question), e the associated evidence, and a the answer generated by a language model. The task is to estimate a reliability score $s^{(d)}$ for dataset d, defined as:

$$s^{(d)} = \mathcal{C}^{(d)}\left(w_1^{(d)} f_{\text{judge}}(q,a) + w_2^{(d)} f_{\text{NLI}}(\tilde{e}^{(d)},a) + w_3^{(d)} f_{\text{consistency}}(\{a_i\}_{i=1}^{N}) - \lambda^{(d)} f_{\text{contra}}(\tilde{e}^{(d)},a)\right) \tag{1}$$

where,

$$\tilde{e}^{(d)} = \begin{cases} e, & \text{if supporting evidence or context is available in dataset } d \\ \varnothing, & \text{otherwise} \end{cases} \tag{2}$$

where f_{judge} denotes a semantic judgement signal obtained from an external language model evaluator, f_{NLI} represents a natural language inference-based factual consistency score, and $f_{\text{consistency}}$ measures cross-sample agreement using multiple generated answers. The weights $w_i^{(d)}$ are dataset-dependent coefficients reflecting the relative importance of each signal. In addition, f_contra denotes a contradiction-based penalty derived from NLI contradiction signals. Its contribution may vary across datasets, and is primarily used as an auxiliary filtering signal in datasets without explicit contradiction supervision (e.g., Natural Questions). $\mathcal{C}^{(d)}(\cdot)$ denotes a calibration function implemented using logistic regression (Platt scaling), which maps the aggregated score to a normalized reliability estimate.

Different datasets (e.g., FEVER, TruthfulQA, and Natural Questions) exhibit varying characteristics in terms of supervision signals and answer formats; therefore, dataset-specific weights and calibration functions are introduced to improve generalization and stability. Although a unified formulation is presented for clarity, not all components are equally applicable across datasets. In particular, certain signals (e.g., contradiction-based penalties or evidence-conditioned scoring) may be omitted or adapted when explicit grounding is unavailable (e.g., TruthfulQA) or only partially applicable (e.g., Natural Questions). This design reflects a dataset-aware implementation in which the general scoring framework is flexibly instantiated according to task-specific constraints.

The score $s^{(d)}$ represents a reliability measure indicating the degree of factual correctness of the answer. A higher score corresponds to higher confidence in factual correctness, while a lower score indicates a higher risk of hallucination.

Based on this score, a decision function is defined to classify the answer as reliable or hallucinated using a threshold τ, resulting in a binary decision:

$$\hat{y} = \begin{cases} \text{reliable}, & \text{if } s \geq \tau \\ \text{hallucinated}, & \text{otherwise} \end{cases} \tag{3}$$

The threshold is tuned on validation data and fixed during test-time evaluation. In addition to this binary classification, the score is further used in a multi-threshold decision mechanism to guide subsequent mitigation actions, enabling different levels of intervention depending on the estimated reliability.

While this decision mechanism follows a unified structure across datasets, its behaviour is dataset-dependent, as different grounding settings and answer formats (e.g., evidence-based verification in FEVER, multi-reference answers in TruthfulQA, and span-based answers in Natural Questions) influence the effectiveness of different intervention strategies.

Unlike conventional approaches that treat hallucination detection as a standalone post-hoc classification problem, this project extends the formulation to a closed-loop framework.

Specifically, the detection score is further used to guide a mitigation function:

$$a' = g^{(d)}(q, \tilde{e}^{(d)}, a, s) \tag{4}$$

where a' is a refined or alternative answer generated to reduce hallucination risk.

In practice, the mitigation function $g(\cdot)$ is implemented as a structured, detector-conditioned decision process based on the reliability score s. This enables adaptive intervention strategies at different confidence levels, ensuring consistency between detection and mitigation, as both stages are governed by the same multi-signal scoring mechanism. Depending on the estimated score, different levels of intervention are applied, including answer rewriting, candidate generation, filtering based on factual consistency, and reranking. In particular, multiple candidate answers are generated and filtered using NLI-based consistency and contradiction-aware constraints, including both detector-level and auxiliary contradiction filtering, followed by label-consistent group-based reranking to select the most reliable answer. In cases where the answer remains uncertain, the system may abstain from producing a definitive response when the reliability score falls below a predefined threshold.

This design enables the system not only to identify unreliable outputs but also to actively correct them, forming a closed-loop pipeline that integrates detection and mitigation. Although the overall mitigation framework is unified, its behaviour varies across datasets due to differences in grounding signals and answer formats, such as evidence-based verification in FEVER, multi-reference evaluation in TruthfulQA, and span-based extraction in Natural Questions.

Overall, this formulation defines hallucination detection, estimation and mitigation as a unified, score-driven process that integrates continuous reliability estimation, binary classification, and adaptive intervention. By combining multi-signal scoring with a structured decision mechanism, the framework provides a principled foundation for improving both detection robustness and mitigation effectiveness.

Building on this formulation, the next section presents the overall system architecture, illustrating how the proposed components are integrated into a complete hallucination detection and mitigation pipeline.

4.2 System Overview

This section presents an overview of the proposed framework for hallucination detection, estimation, and mitigation in large language models. The system is designed as a closed-loop pipeline that integrates answer generation, reliability estimation, and corrective intervention into a unified process. Unlike conventional approaches that treat hallucination detection as a post-hoc classification task, the proposed framework uses reliability estimation to actively guide the generation process.

Given an input claim or question q and an optional evidence context e (if available), the system first generates an initial answer a using a large language model. This answer is then analysed by a multi-signal detection module, which estimates its factual reliability from multiple complementary perspectives. These signals include semantic judgement, factual consistency, contradiction-aware signals, and cross-sample consistency, allowing the system to capture different types of hallucination behaviour.

The outputs of these signals are aggregated into a unified reliability score through a weighted combination and calibration process. This score provides a continuous estimate of factual correctness and serves as the control signal for subsequent decision-making.

Based on the estimated reliability, a decision module determines whether the answer should be accepted or further refined. High-confidence answers are preserved, while lower-confidence answers trigger mitigation procedures of varying intensity. This results in a score-conditioned intervention mechanism that adapts the level of correction to the estimated hallucination risk.

When mitigation is required, the system generates multiple candidate answers and applies a structured selection process to identify the most reliable one. This process includes consistency-based filtering and reliability-aware reranking. The filtering criteria are dataset-aware and may rely on evidence when available (e.g., FEVER and Natural Questions), or on internal consistency signals when external grounding is not provided (e.g., TruthfulQA). If no candidate meets the required reliability criteria, the system may abstain from producing a final answer.

Overall, this design forms a closed-loop framework in which detection and mitigation are tightly coupled, enabling the system not only to identify hallucinated outputs but also to actively improve them while preserving answer quality.

4.3 Datasets

To comprehensively evaluate the effectiveness, robustness, and generalisation ability of the proposed hallucination detection, estimation and mitigation framework, three widely used benchmark datasets are employed in this study: TruthfulQA, FEVER, and Natural Questions (NQ). These datasets are selected to represent different types of question-answering scenarios and hallucination characteristics, ranging from adversarial questions to evidence-based verification and open-domain extractive QA.

4.3.1 TruthfulQA

TruthfulQA is designed to evaluate whether large language models produce truthful answers to questions that are prone to common misconceptions. Unlike standard QA datasets, many questions in TruthfulQA are intentionally constructed such that naive or commonly memorised responses are incorrect.

This dataset is particularly suitable for hallucination evaluation because:

- It explicitly targets model-generated misinformation
- It includes multiple correct and incorrect reference answers
- It requires semantic understanding rather than surface matching

In this project, TruthfulQA is used to evaluate the system's ability to detect and mitigate hallucinations in open-ended generative settings, where answers are not constrained by context.

In the experimental setup of this project, TruthfulQA is used in a generation-only setting without external context. This makes TruthfulQA particularly suitable for evaluating hallucination in the absence of explicit grounding signals. The model is prompted to produce answers directly based on its internal knowledge.

For evaluation, the generated answers are compared against both correct and incorrect reference answers provided in the dataset. A multi-signal hallucination detection mechanism is then applied, where semantic judgement, NLI-based consistency signals, and self-consistency signals are combined to estimate the factual reliability of the answer and identify potential hallucinations. Contradiction signals derived from incorrect reference answers are additionally used to penalise misleading or factually inconsistent outputs. Since no external evidence is available, NLI-based signals are computed using reference answers (both correct and incorrect) rather than grounded evidence.

This setup enables the assessment of hallucination behaviour in purely generative scenarios, where no explicit grounding is available and the model must rely entirely on its parametric knowledge.

4.3.2 FEVER

FEVER is a fact verification dataset consisting of claims labelled as SUPPORTED, REFUTED, or NOT ENOUGH INFORMATION, along with corresponding evidence retrieved from Wikipedia.

FEVER differs from standard QA datasets in that it focuses on claim verification rather than answer generation. It provides structured evidence that allows explicit reasoning about factual consistency.

This dataset is used in this project to:

- Evaluate hallucination detection under evidence-based settings
- Leverage explicit evidence to support mitigation
- Test the effectiveness of contradiction detection mechanisms

In this project, each FEVER instance is treated as a verification task, where the input claim is used as the question and the annotated evidence is used as grounding context. The model is encouraged to generate a response that is consistent with the provided evidence.

For evaluation, the generated answer is compared against the ground truth label (SUPPORTED, REFUTED, or NOT ENOUGH INFORMATION). In addition, the evidence is used as a reference for entailment-based verification, where natural language inference (NLI) models are applied to determine whether the generated answer is supported or contradicted by the evidence.

This setup enables a grounded evaluation of hallucination, where factual correctness is explicitly tied to external evidence rather than relying solely on parametric knowledge. It also allows the proposed mitigation framework to leverage evidence during answer correction, improving factual consistency through external grounding. This also allows the integration of contradiction-aware filtering mechanisms during mitigation.

4.3.3 Natural Questions(NQ)

Natural Questions is a large-scale dataset consisting of real user queries paired with answers from Wikipedia documents. Answers are typically short spans extracted directly from the source text.

Compared to TruthfulQA and FEVER, Natural Questions represents a more realistic and practical QA scenario, where:

- Questions are naturally occurring
- Answers are extractive and context-dependent
- Long documents introduce noise and complexity

In this project, Natural Questions is implemented in an extractive QA setting, where a context window is constructed around the annotated answer span from the original document. Instead of allowing free-form generation, the model is prompted to produce answers that are constrained via prompting to produce answers that correspond to spans in the provided context.

To achieve this, a constrained prompting strategy is adopted, requiring the model to output a contiguous span that exactly appears in the context. This reduces the likelihood of hallucination by limiting the model's output space to grounded evidence.

For evaluation, the generated answer is compared against the reference short answer using exact match and token-level overlap metrics. In addition, hallucination detection signals, including entailment-based signals are used as auxiliary consistency checks, while contradiction signals play a limited role due to the lack of explicit contradiction supervision.

This setup allows the analysis of hallucination behaviour under partially grounded conditions, where relevant information is available but must be correctly identified within noisy and lengthy documents. This setting emphasises the importance of precise span identification and reduces reliance on free-form generation.

The combination of these three datasets enables a comprehensive evaluation across diverse settings: (1) adversarial generation without grounding (TruthfulQA), (2) evidence-based verification (FEVER), and (3) extractive QA with local context (Natural Questions). This design ensures that the proposed framework is not overfitted to a single task type and provides insights into its stability and generalisation across different hallucination scenarios.

To address the inherent stochasticity of large language models, a controlled experimental protocol is adopted. For each dataset, fixed subsets of samples are constructed, and multiple independent runs are conducted using different random seeds. This enables the measurement of both average performance and variability, thereby providing a more reliable and reproducible evaluation of hallucination detection and mitigation performance.

In terms of data preparation, all datasets are obtained from the HuggingFace Datasets library and subsequently preprocessed into local JSON files to facilitate efficient loading and reproducibility.

Due to the large scale of the FEVER and Natural Questions datasets, it is computationally expensive to perform full-scale experiments, particularly under the multi-run evaluation setting. Therefore, subset construction is applied in a two-stage manner.

First, intermediate data pools are created. For the FEVER dataset, stratified sampling is applied by grouping the data according to labels (SUPPORTED, REFUTED, NOT ENOUGH INFORMATION) and randomly selecting an equal number of samples from each class, resulting in a balanced subset of 3,000 instances. For the Natural Questions dataset, simple random sampling is applied to select 2,000 instances, preserving the natural distribution of question types and answer structures.

Second, during experimentation, smaller subsets are further sampled from these intermediate data pools in each run. This enables a Monte Carlo-style evaluation protocol, where different subsets are used across runs to assess both performance and stability.

Overall, this design strikes a balance between computational efficiency and experimental validity, while preserving the diversity and complexity of the original datasets and ensuring consistent and reproducible evaluation.

4.4 Initial Answer Generation

4.4.1 Role of Initial Answer Generation in the Pipeline

The initial answer generation stage serves as the foundation of the entire hallucination detection and mitigation pipeline. Given an input question, the system generates one or more candidate answers using a large language model under a controlled setting. However, the generation strategy is adapted according to the characteristics of each dataset to ensure both validity and consistency.

For datasets with local contextual grounding (e.g., Natural Questions), answer generation is guided by a context window constructed from the source document. The model is prompted to produce answers that correspond to spans in the provided context, encouraging extractive behaviour without strictly enforcing hard constraints.

In contrast, for datasets without explicit grounding (e.g., TruthfulQA), answers are generated in an open-ended setting without external context. This allows the system to evaluate hallucination under unconstrained generation conditions, where the model relies primarily on its internal knowledge.

For evidence-based verification datasets (e.g., FEVER), the generation process is aligned with the claim verification setting. Although answers are generated in a free-form manner, the availability of structured evidence allows downstream modules to perform explicit verification using NLI-based entailment and contradiction signals, which also support contradiction-aware filtering during mitigation.

Furthermore, instead of producing a single answer, multiple candidate answers are generated for each question across all datasets. In this work, three candidate answers are generated per question through independent generation runs. This design enables the system to capture variability in model outputs and supports downstream self-consistency estimation via NLI-based agreement scoring across generated answers..

The choice of generating five samples reflects a trade-off between computational efficiency and estimation reliability. While multi-sample approaches improve robustness by exposing inconsistencies across outputs, increasing the number of samples leads to diminishing returns and higher computational cost. Therefore, a fixed sample size of five is adopted as a practical and effective setting.

As a result, the generation process is treated as a controlled and analyzable intermediate step, rather than a black-box component.

This design explicitly adapts the generation strategy to different task settings and constrains the generation process where appropriate. As a result, the system ensures that the produced answers are both suitable for downstream multi-signal evaluation and representative of the model's behaviour under diverse conditions.

4.4.2 Context Construction

For datasets that provide supporting information (e.g., Natural Questions and FEVER), a context construction step is applied to create an evidence-grounded input for answer generation. However, the construction strategy differs depending on the structure of the dataset.

For **Natural Questions**, where answers correspond to spans within long documents, a local context window is constructed. Specifically, given a reference answer extracted from the dataset (e.g., short answer text or annotated token spans), the system attempts to locate the corresponding position within the source document. This is achieved by matching either the provided answer text or token span annotations to the document tokens. Once the span is identified, a local context window is constructed by selecting a fixed number of tokens before and after the matched position. In this work, a fixed-size context window is constructed by selecting a limited number of tokens before and after the matched position (e.g., a symmetric window around the answer span).

This design aims to ensure that the constructed context:

- Contains sufficient relevant information to support answer extraction
- Reduces irrelevant noise from long documents
- Preserves locality, which is important for extractive QA settings

If the reference span cannot be reliably located within the document, a fallback strategy is applied. In this case, the first 600 tokens of the document are used as a proxy context, ensuring that a valid input is always available for generation.

For **FEVER**, where each claim is associated with annotated evidence sentences, the provided evidence is directly used as the context. In practice, a simplified strategy is adopted where a representative evidence sentence is selected from the annotated evidence set, providing a concise context for verification. Unlike Natural Questions, no span-based window construction is required, as the dataset already provides concise and structured evidence supporting or refuting the claim. This enables direct factual verification without additional context extraction.

Unlike retrieval-augmented approaches that depend on external search systems, both strategies rely solely on dataset-provided information. This ensures that the context construction process remains deterministic and consistent across experiments, thereby improving reproducibility and eliminating variability introduced by retrieval components.

However, it should be noted that the Natural Questions setting constructs context based on oracle reference spans, which may not fully reflect real-world retrieval settings where evidence must be dynamically retrieved. While this design improves internal validity by ensuring the presence of relevant evidence, it may still introduce challenges such as incomplete context coverage or residual noise, particularly in long documents. These limitations are further discussed in the evaluation section.

4.4.3 Extractive Answer Generation via Prompt Engineering

Answer generation is implemented using a prompt-engineering-based extractive strategy, where the language model is guided to select answers from the provided context. Rather than allowing fully unconstrained natural language generation, the prompt encourages extractive behaviour at the token level.

The prompt is designed to simulate a constrained extractive QA setting by imposing the following constraints:

- The answer is encouraged to be a contiguous span appearing in the context
- Paraphrasing or reformulation is discouraged
- The model is instructed to avoid introducing external knowledge beyond the given context
- The output follows a predefined format (i.e., `FINAL_ANSWER:`)

In addition, the prompt explicitly discourages non-informative responses such as "I don't know", encouraging the model to produce a valid span whenever possible. This constraint is critical for maintaining consistency with downstream evaluation and mitigation components.

This prompt design effectively transforms the generative model into a constrained extraction mechanism, allowing controlled observation of model behaviour under evidence-grounded conditions. Unlike traditional extractive QA models trained for span prediction, this approach leverages prompt-based control to achieve similar behaviour without additional training.

Such a design enforces strict output constraints, reduces variability in answer formats and ensures compatibility with subsequent hallucination detection and scoring modules. However, it should be noted that these constraints are enforced at the prompt level rather than through hard decoding constraints, and therefore the behaviour remains probabilistic.

4.4.4 Controlled Generation Strategy

To improve reproducibility while maintaining task-appropriate behaviour, controlled decoding strategies are applied in a dataset-specific manner rather than enforcing a single global configuration.

TruthfulQA.
For TruthfulQA, the primary objective is to expose potential hallucinations and evaluate model consistency. Therefore, moderately stochastic decoding is adopted during initial answer generation (e.g., temperature ≈ 0.3, top-$p \approx 0.9$) to encourage diversity across multiple sampled answers. This variability is essential for enabling self-consistency analysis and improving the robustness of hallucination detection.

During mitigation, lower temperature settings are used to suppress hallucinations. Mild rewriting adopts a moderate temperature (e.g., ≈ 0.3) to allow limited correction, while strict rewriting uses near-deterministic decoding (e.g., temperature ≈ 0.1–0.2) to enforce conservative and factual outputs.

FEVER.
For FEVER, where ground-truth evidence is explicitly provided, deterministic decoding is preferred. The model is configured with temperature $= 0$ and top-$p = 1.0$ to ensure that outputs are consistent and closely aligned with the provided evidence. This reduces unnecessary variability and helps the model focus on evidence-grounded reasoning rather than generating diverse interpretations.

Natural Questions (NQ).
For Natural Questions, which follows an extractive question answering paradigm, deterministic decoding is also adopted (temperature $= 0$). Since answers are expected to align closely with short reference spans, reducing stochasticity helps improve answer precision and consistency, especially when combined with extractive mitigation strategies.

Deterministic Components.

For evaluation modules such as LLM-as-a-judge, fully deterministic decoding (temperature = 0) is consistently applied across all datasets. This ensures stable and reproducible judgments, avoiding additional noise introduced by stochastic decoding.

Residual Variability

Despite the use of deterministic decoding in several components, the overall system remains subject to residual variability. As discussed in Section 2.5, large language models are inherently stochastic due to factors such as implementation-level nondeterminism, backend execution differences (e.g., GPU parallelism), and subtle variations in inference conditions [38].

In addition, the generation pipeline introduces further variability through answer post-processing, fallback strategies, and retry mechanisms when invalid outputs are produced. To account for this, deterministic decoding is complemented with multi-sample answer generation (Section 4.4.5), allowing the system to capture and analyse remaining variability rather than assuming strict determinism.

4.4.5 Multi-Sample Answer Generation

To capture residual variability in model outputs and support self-consistency-based evaluation, multiple answers are generated for each input question. However, the role of multi-sample generation differs across datasets, and is therefore applied in a task-dependent manner.

TruthfulQA.

For TruthfulQA, multi-sample generation plays a central role in hallucination detection. Given a question q, the model generates a set of candidate answers:

$$A = \{a_1, a_2, \dots, a_n\}, \quad n = 5$$

Each answer is produced through an independent generation process under moderately stochastic decoding (as described in Section **??**). The resulting diversity allows the system to perform self-consistency estimation, where agreement between answers is measured using a SelfCheckGPT-style NLI-based approach.

This design enables the detection of unstable or contradictory responses, which are strong indicators of hallucination.

FEVER.

For FEVER, the task is evidence-based claim verification, and answers are expected to align with provided evidence. As a result, multi-sample generation is not primarily used for diversity, but rather for improving robustness against residual system-level variability. Candidate answers are generated under deterministic decoding (temperature = 0), and multiple samples help ensure that evaluation is not biased by a single potentially unstable output.

Natural Questions (NQ).

For Natural Questions, which follows an extractive question answering paradigm, multi-sample generation is used more conservatively. Since answers are expected to match short reference spans, excessive diversity is undesirable. Therefore, multiple samples are mainly used to improve robustness and reduce sensitivity to occasional generation noise, rather than to encourage variation.

Caching and Reuse.

To improve computational efficiency and ensure reproducibility, generated answers are cached and reused across experiments. Once a set of answers is generated for a given question, it is stored and subsequently reused to avoid redundant model calls.

Role in Evaluation.

The resulting set of candidate answers serves two main purposes:

- **Self-consistency estimation:** agreement between answers is measured using an NLI-based approach inspired by SelfCheckGPT, allowing the system to quantify internal consistency.
- **Robust evaluation:** aggregating multiple outputs reduces sensitivity to single-instance randomness and provides a more stable estimate of model behaviour.

This dataset-aware design ensures that multi-sample generation is used to capture meaningful variability when beneficial (e.g., hallucination detection in TruthfulQA), while avoiding unnecessary diversity in tasks requiring high precision and grounding (e.g., FEVER and NQ). This approach aligns with prior work showing that multi-sample generation improves the reliability of LLM evaluation and facilitates more robust detection of inconsistencies in generated content [10].

4.4.6 Output Parsing and Normalisation

The raw outputs produced by large language models may contain additional text beyond the intended answer span, such as explanations, formatting artifacts, or incomplete responses. To ensure reliable and consistent downstream evaluation, a structured parsing and normalisation procedure is applied as a unified post-processing step. While this procedure is shared across all datasets, its functional role varies depending on the task setting.

Answer Extraction.

The system first attempts to extract a well-defined answer span using a predefined output pattern. Specifically, if the model output contains a segment prefixed with `FINAL_ANSWER:`, the content following this marker is extracted. If this pattern is not present, a fallback strategy is applied, where only the first line of the output is retained as the candidate answer. This design enforces a consistent single-answer format while remaining robust to deviations in model output structure.

Cleaning.

After extraction, lightweight cleaning operations are performed to remove formatting artifacts such as quotation marks or extraneous surrounding text. Given the variability in output formats across datasets (e.g., free-form answers in TruthfulQA, categorical labels in FEVER, and span-based answers in Natural Questions), the cleaning process is intentionally minimal to avoid removing semantically meaningful content.

Normalisation.

The extracted answer is further normalised to ensure consistency across samples. This includes:

- Collapsing multiple whitespace characters into a single space
- Removing trailing punctuation (e.g., periods)

- Eliminating residual formatting artifacts

These steps standardise answer representations without altering their semantic content.

Output Validation.

To improve robustness, an additional validation step is applied to filter out invalid or non-informative responses. Outputs that are empty, malformed, or trivial are treated as invalid and handled through fallback mechanisms during the generation stage. This prevents such cases from propagating into downstream evaluation.

Dataset-Specific Considerations.

Although a unified parsing procedure is applied, its relative importance differs across datasets due to variations in output structure and evaluation criteria:

- **TruthfulQA:** Outputs are open-ended and may include explanatory or speculative text. Parsing primarily serves to isolate concise answer spans from potentially verbose generations, ensuring comparability across samples.

- **FEVER:** Outputs are short categorical labels (e.g., `SUPPORTED`, `REFUTED`, `NOT ENOUGH INFO`). Parsing mainly ensures consistent extraction and formatting of these labels, reducing errors caused by minor formatting variations.

- **Natural Questions:** Outputs correspond to extractive spans that are evaluated using lexical matching (e.g., token-level F1). In this setting, parsing and normalisation are particularly critical, as even minor formatting inconsistencies can significantly affect evaluation results.

Summary.

Overall, this procedure standardises model outputs into a consistent and comparable format while preserving dataset-specific characteristics. This is essential for downstream evaluation, including lexical matching and NLI-based scoring, and ensures that comparisons across datasets remain fair, robust, and reproducible.

4.4.7 Invalid Output Handling and Fallback Strategy

Despite structured prompting and parsing, language models may still produce invalid or unusable outputs. Typical failure cases include:

- Empty responses
- Non-informative or trivial outputs (e.g., "unknown")
- Outputs that do not conform to the expected format

To ensure robustness, a lightweight invalid output handling procedure is applied. Invalid outputs are detected using simple heuristic rules based on content validity (e.g., empty, overly short, or semantically uninformative responses). The objective is to ensure that all generated samples remain usable for downstream evaluation, without altering the semantic content of valid outputs.

Dataset-Specific Handling.
Rather than applying a unified fallback mechanism, the system adopts dataset-aware handling strategies aligned with task requirements:

- **TruthfulQA:** As an open-ended generation task without external grounding, TruthfulQA prioritises preserving the model's natural output behaviour. Invalid outputs are handled using a parsing-based fallback mechanism: the system first attempts to extract the answer using the `FINAL_ANSWER:` pattern, and if unavailable, retains the raw output after minimal trimming.

  No correction or rewriting is applied at this stage, ensuring that evaluation reflects the model's original behaviour. Responses corresponding to abstention (e.g., "I do not know") are normalised into a consistent representation, enabling systematic treatment during downstream scoring.

- **FEVER:** For FEVER, where outputs correspond to categorical labels or short factual statements, invalid outputs are handled conservatively. Parsing and normalisation are applied to ensure consistent formatting, but no correction or rewriting is performed. This preserves the integrity of the model's predictions for subsequent evidence-based evaluation.

- **Natural Questions:** In the extractive QA setting of Natural Questions, stricter constraints are required. The system enforces extractive behaviour at multiple levels: prompt design restricts outputs to contiguous spans from the context, and parsing ensures clean and standardised answer formatting.

For invalid or unusable outputs, a fallback extraction mechanism is applied, where a plausible answer span is directly selected from the context. Candidate answers are further validated to ensure that they correspond to spans explicitly present in the context, preventing hallucinated or out-of-context responses.

Design Rationale.

This design ensures that all generated samples remain valid and comparable across datasets while respecting task-specific constraints. Importantly, invalid output handling is limited to structural correction (e.g., extraction, normalisation, or fallback span selection) and does not introduce semantic modifications at the initial generation stage. Semantic corrections are only introduced in later mitigation stages, preserving a clear separation between generation, validation, and correction.

4.4.8 Justification of Design Choices

The design of the initial answer generation module is guided by the need to balance factual grounding, controllability, stability, and compatibility with downstream hallucination detection. Rather than treating generation as an isolated component, it is explicitly designed to support a multi-stage, evaluation-driven pipeline.

(1) Grounding vs. hallucination risk.

To reduce hallucination, grounding is incorporated where it is both available and compatible with the task formulation. In extractive question answering settings such as *Natural Questions*, the model is constrained through prompt design to generate answers as spans grounded in the provided context. This substantially reduces the risk of unsupported or fabricated content.

In contrast, for datasets without explicit grounding during generation (e.g., *TruthfulQA* and *FEVER*), strict extractive constraints are not imposed. Although *FEVER* provides evidence, enforcing extractive grounding at generation time may conflict with its classification-oriented formulation. Instead, hallucination risks are handled in downstream evaluation through NLI-based verification and scoring.

This dataset-aware strategy ensures that grounding is applied when beneficial, while preserving task-appropriate generation behaviour in other settings.

(2) Prompt-based control.

The system adopts prompt engineering rather than model fine-tuning as the primary control mechanism. Task-specific prompts enforce structural constraints (e.g., extractive spans or standardised output formats such as `FINAL_ANSWER:`), enabling consistent behaviour across heterogeneous datasets.

This approach avoids additional training overhead, improves reproducibility, and allows rapid adaptation to different task settings, making it a practical and scalable design choice for LLM-based systems.

(3) Stability and controlled variability.

Low-temperature decoding is used to improve output stability by reducing stochastic variation. However, deterministic decoding alone does not eliminate variability in large language models. To capture residual uncertainty, multiple answers are generated for each question and preserved for downstream analysis.

This design enables self-consistency estimation and uncertainty modelling, treating variability as a signal rather than noise, and contributing to more robust evaluation.

(4) Compatibility with detection signals.

The generation module is explicitly designed to align with the downstream multi-signal hallucination detection framework. In particular:

- Structured outputs enable reliable parsing and normalisation
- Multiple sampled answers support self-consistency scoring
- Clean outputs facilitate NLI-based factual verification
- Standardised formats ensure consistent LLM-as-a-judge evaluation

This tight integration minimises additional processing overhead and ensures that detection signals can be applied reliably and consistently.

(5) Robustness through controlled fallback.

To ensure robustness, a lightweight fallback strategy is applied to handle invalid outputs (e.g., empty or malformed responses). These mechanisms enforce structural validity without introducing semantic modifications.

In extractive settings, fallback operations select plausible spans directly from the context. In non-extractive settings, outputs are normalised and retained. This conservative design avoids evaluation bias while ensuring that all samples remain usable.

Summary.

Overall, the generation module is designed as an integral part of a hallucination-aware pipeline. By combining dataset-aware grounding, prompt-based control, controlled variability, and compatibility with downstream signals, it provides a stable and principled foundation for subsequent detection and mitigation stages.

4.4.9 Summary

In summary, the initial answer generation module adopts a controlled, dataset-aware, and multi-sample generation strategy. Extractive prompting is applied where compatible with the task formulation (e.g., *Natural Questions*), while non-extractive generation is used for open-ended and classification-style tasks such as *TruthfulQA* and *FEVER*.

By combining context grounding (where applicable), prompt-based constraints, low-temperature decoding, and lightweight post-processing, the system produces structured and valid candidate answers suitable for downstream hallucination detection and mitigation.

This design balances accuracy, robustness, and consistency, and provides a reliable foundation for the subsequent multi-signal evaluation framework.

4.5 Hallucination Detection Model

This section introduces the proposed hallucination detection model, which uses a multi-signal method to estimate the factual reliability of generating answers. Unlike the single-signal method, which usually relies on confidence scores or isolated metrics, the proposed model integrates multiple complementary signals to capture different dimensions of hallucination behavior.

4.5.1 Overview of Detection Framework

4.5.2 Semantic Judgement Signal

4.5.3 NLI-based Factual Consistency

4.5.4 Self-consistency Signal

4.5.5 Score Aggregation

4.6 Mitigation Strategy (Closed-loop)

4.7 Thresholding & Calibration

4.8 Evaluation Protocol

Chapter 5

Implementation (Technical Achievement)

Chapter 6

Experimental Setup

Chapter 7

Results & Discussion

Chapter 8

Project Management

Chapter 9

Reflection and Future Work

9.1 Experimental Results

9.2 Analysis of Results

9.3 Discussion in Wider Context

9.4 Limitations

Chapter 10

Conclusion

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